CLINICAL TRIAL FAILURES · MISSED ENDPOINTS · RESPIRATORY PHASE 2,3 · STARTS 2015–2024

IL-5 / IL-5R: 5 of 12 completed trials missed their primary endpoint

5 completed trials across 3 sponsors and 3 molecules ran to the end and missed their primary endpoint on the sponsor's own posted result; 7 met it. Only 2 of 34 closed trials were stopped early for efficacy or safety, which is why this class is read through its completed trials rather than a stop rate. Most completed trials post nothing readable, so the count is a floor.

5 of 12
completed trials with a readable primary result missed it · 7 met it

3 / 3
sponsors / molecules behind the misses

5.9%
stopped early: 2 of 34 closed trials (95% CI 1.6%–19.1%)

81.0%
of 42 trials have closed · 8 still open or unresolved

5 of 12 **Completed and missed the primary endpoint.** Of 30 completed trials in this segment, 12 posted a primary result on ClinicalTrials.gov that can be read — the sponsor's own comparison held to the sponsor's own threshold, or its statement that the endpoint was missed: 5 missed and 7 met. Counted separately and never in the rate above; a floor, not a rate. 3 of the 28 completed trials that finished more than a year ago have posted no results at all — a gap in what can be known, not a sign of failure.

The completed trials that missed

Trial	Ph	Sponsor	Experimental drugs	Started	What the posted result says
NCT04075331	3	University of Leicester	MEPOLIZUMAB	2020-09	Time From Randomisation to Next Hospital Readmission or Death (All Cause) — p 0.811, no threshold stated; 0.1 or above, so not significant under any conventional threshold. Hazard Ratio (HR) 0.96 (95% CI 0.7 to 1.32). Method: Log Rank.
NCT02869438	3	AstraZeneca	BENRALIZUMAB	2016-11	Change From Baseline (Visit 4) to Day 28 (Visit 8), Day 56 (Visit 9), and Day 84 (Visit 10) in Pre-BD FEV1 — p 0.7747, no threshold stated; 0.1 or above, so not significant under any conventional threshold. Mean Difference (Final Values) 0.013 (95% CI -0.077 to 0.104). Method: Mixed Models Analysis.
NCT02821416	3	AstraZeneca	BENRALIZUMAB	2016-10	Maximal Percentage Decrease in Forced Expiratory Volume in 1 Second 3-7 Hours Post Allergen Challenge — p 0.363, no threshold stated; 0.1 or above, so not significant under any conventional threshold. Median Difference (Final Values) 2.535 (95% CI -3.045 to 8.116). Method: Mixed Models Analysis.
NCT02501629	3	Teva Branded Pharmaceutical Products R&D, Inc.	RESLIZUMAB	2015-09	Number of Participants With Reduction In Daily Oral Corticosteroid (OCS) Dose During Weeks 20-24 As Compared to the Optimized Dose At Baseline — p 0.468, no threshold stated; 0.1 or above, so not significant under any conventional threshold. Odds Ratio (OR) 1.23 (95% CI 0.702 to 2.157). Method: proportional odds model.
NCT02452190	3	Teva Branded Pharmaceutical Products R&D, Inc.	RESLIZUMAB	2015-09	Number of Clinical Asthma Exacerbations (CAEs) During 52 Weeks of Treatment — p 0.194 against the sponsor's stated threshold of 0.05: not significant. CAE rate ratio (reslizumab vs placebo) 0.79 (95% CI 0.562 to 1.124). Method: Chi-squared.

How this compares

IL-5 / IL-5R



5.9%

2/34 closed

all respiratory trials with a resolved drug target



4.2%

35/840 closed

All respiratory Phase 2,3



4.2%

82/1939 closed

What is in this class: 5 distinct experimental drugs across 42 trials (antibody 42, small molecule 1). Most tested: BENRALIZUMAB (20), MEPOLIZUMAB (10), RESLIZUMAB (7), DEPEMOKIMAB (5), MOMETASONE FUROATE (1). A class groups drugs by what they act on; check that the grouping is one you would make before reading the rate as a property of the mechanism.

How much does this rest on one decision? The 2 stops came from 2 sponsor-asset programmes across 2 sponsors; the largest (Teva Branded Pharmaceutical Products R&D, Inc.) contributed 1; removing that programme's trials from both sides leaves 3.2%. Screened alongside 13 other segments, the rate is within what screening that many classes could produce (q=1); the molecules and trials that failed here are the record either way. One further closed trial here was terminated with no cause recorded in the registry; if it were biological the rate would be 8.8%. Read it as a screen worth checking against the underlying trials.

Bars show the share of closed trials stopped for biological reasons. Confidence intervals overlap where sample sizes are small — read the counts, not just the bars.

By start cohort

Trials started	Stops	Closed	Rate	95% CI
2015–2017	1	19	5.3%	0.9%–24.6%
2018–2020	1	8	12.5%	2.2%–47.1%
2021–2024	0	7	0.0%	0.0%–35.4%

Where the stops sit

Efficacy / futility: 2.
Sponsors with most stops: AstraZeneca (1), Teva Branded Pharmaceutical Products R&D, Inc. (1).

The stopped trials

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT04157335	3	AstraZeneca AZN	BENRALIZUMAB	2024-08	Efficacy	The study has a double-blind period and an open-label period. The study did not meet its primary endpoints after the primary LSLV and the decision was taken to terminate the study early.
NCT03052725	3	Teva Branded Pharmaceutical Products R&D, Inc.	RESLIZUMAB	2018-02	Efficacy	Parent Studies didn't meet their primary endpoint so study was terminated.

Method
Denominator: ClinicalTrials.gov interventional Phase 2,3 respiratory trials started 2015–2024 that have closed (completed or terminated). Numerator: terminated trials whose registry stop reason is classified as biological (efficacy, safety or benefit–risk) by Classification V2 — held-out precision 95.5%, recall 95.3% (n=600). Drugs are linked to ChEMBL and the NCI Thesaurus; 41% of industry respiratory trials in this window carry a resolved drug target, and a trial without one cannot enter a mechanism class. Intervals are Wilson 95%.

Limits
Not a failure rate: completed trials that missed their endpoints are not in the rate. They are counted separately where the sponsor posted a primary result that can be read — most did not, so that count is a floor — and programmes discontinued after a completed trial do not appear. Stop reasons are sponsor-reported. This is a closed-trial proportion, not a time-to-event analysis: only 81.0% of trials in this segment have closed, and a trial that stops early enters the denominator sooner than one that runs to completion, which can inflate the rate in immature segments. Recent cohorts have fewer closed trials, so their rates are less stable. Research signals, not clinical or investment advice.

The whole cohort, trial by trial. The diligence report names every trial behind these figures — the ones still running and the ones that completed and missed their primary endpoint included — and places your molecule against each one that failed. Free during the beta, in exchange for a work email, delivered immediately: **clinicaltrialfailures.com/asset-check**

Sources: ClinicalTrials.gov (NLM); ChEMBL (EMBL-EBI, CC BY-SA 3.0); NCI Thesaurus (NCI); RxNorm/RxClass (NLM); SEC EDGAR. Classification, linkage and rates by Clinical Trial Failures. Rebuilt weekly; this brief covers trials started 2015–2024.